

Simulink Implementation of Task-Space Trajectory
Tracking with Motor-Side Decoupled
Computed-Torque Control

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Chapter 1

Simulink Implementation of Task-Space Trajectory Tracking with Motor-Side Decoupled Computed-Torque Control

1.1 Objective

The objective of this work is to implement a complete Simulink model for task-space trajectory tracking of a 2-DOF spherical robot. The robot motion is planned in task space using the orientation variables

$$\boldsymbol{\phi} = \begin{bmatrix} \phi_1 \\ \phi_2 \end{bmatrix}, \quad (1.1)$$

where ϕ_1 is the azimuth angle and ϕ_2 is the tilt angle of the end-effector direction.

The desired task-space trajectory is converted into joint-space position, velocity, and acceleration using inverse kinematics, the Jacobian velocity relation, and the second-order Jacobian relation. Since the actuator dynamics are modeled on the motor side, the joint-side references are converted to motor-side references using the gearbox ratio.

The complete simulation chain is

$$\begin{aligned} \boldsymbol{\phi}_d, \dot{\boldsymbol{\phi}}_d, \ddot{\boldsymbol{\phi}}_d &\rightarrow \text{IK} \rightarrow \boldsymbol{\theta}_{j,d}, \dot{\boldsymbol{\theta}}_{j,d}, \ddot{\boldsymbol{\theta}}_{j,d} \\ &\rightarrow \boldsymbol{\theta}_{m,d}, \dot{\boldsymbol{\theta}}_{m,d}, \ddot{\boldsymbol{\theta}}_{m,d} \rightarrow \boldsymbol{\tau}_m \\ &\rightarrow \boldsymbol{\theta}_m \rightarrow \boldsymbol{\theta}_j \rightarrow \text{DK} \rightarrow \boldsymbol{\phi}_{act}. \end{aligned} \quad (1.2)$$

The aim is to verify that the actual task-space output follows the desired task-space trajectory:

$$\boldsymbol{\phi}_{act} \approx \boldsymbol{\phi}_d. \quad (1.3)$$

1.2 Task-Space Trajectory Generation

The desired orientation trajectory is generated in task space. The final-frame direction vector is defined as

$$\mathbf{d} = \begin{bmatrix} -\cos(\phi_2) \sin(\phi_1) \\ \cos(\phi_2) \cos(\phi_1) \\ \sin(\phi_2) \end{bmatrix}. \quad (1.4)$$

A quintic polynomial is used for each task-space variable because it gives smooth position, velocity, and acceleration profiles.

For one task variable,

$$\phi(t) = a_0 + a_1t + a_2t^2 + a_3t^3 + a_4t^4 + a_5t^5. \quad (1.5)$$

The velocity is

$$\dot{\phi}(t) = a_1 + 2a_2t + 3a_3t^2 + 4a_4t^3 + 5a_5t^4. \quad (1.6)$$

The acceleration is

$$\ddot{\phi}(t) = 2a_2 + 6a_3t + 12a_4t^2 + 20a_5t^3. \quad (1.7)$$

The boundary conditions are

$$\phi(0) = \phi_0, \quad \phi(t_f) = \phi_f, \quad (1.8)$$

$$\dot{\phi}(0) = 0, \quad \dot{\phi}(t_f) = 0, \quad (1.9)$$

$$\ddot{\phi}(0) = 0, \quad \ddot{\phi}(t_f) = 0. \quad (1.10)$$

For the two task-space variables,

$$\boldsymbol{\phi}_d(t) = \begin{bmatrix} \phi_{1d}(t) \\ \phi_{2d}(t) \end{bmatrix}, \quad (1.11)$$

$$\dot{\boldsymbol{\phi}}_d(t) = \begin{bmatrix} \dot{\phi}_{1d}(t) \\ \dot{\phi}_{2d}(t) \end{bmatrix}, \quad (1.12)$$

$$\ddot{\boldsymbol{\phi}}_d(t) = \begin{bmatrix} \ddot{\phi}_{1d}(t) \\ \ddot{\phi}_{2d}(t) \end{bmatrix}. \quad (1.13)$$

These three signals are provided to Simulink using *From Workspace* blocks.

1.3 Inverse Kinematics

The inverse kinematics block receives the desired task-space orientation

$$\boldsymbol{\phi}_d = \begin{bmatrix} \phi_{1d} \\ \phi_{2d} \end{bmatrix} \quad (1.14)$$

and computes the desired joint-side motor angles

$$\boldsymbol{\theta}_{j,d} = \begin{bmatrix} \theta_{1d} \\ \theta_{2d} \end{bmatrix}. \quad (1.15)$$

The desired task-space direction is

$$\boldsymbol{d}_d = \begin{bmatrix} -\cos(\phi_{2d}) \sin(\phi_{1d}) \\ \cos(\phi_{2d}) \cos(\phi_{1d}) \\ \sin(\phi_{2d}) \end{bmatrix}. \quad (1.16)$$

Using the final-frame conversion,

$$\boldsymbol{d} = \begin{bmatrix} -v_1(2) \\ v_1(1) \\ v_1(3) \end{bmatrix}, \quad (1.17)$$

the inverse conversion is

$$\boldsymbol{v} = \begin{bmatrix} d_y \\ -d_x \\ d_z \end{bmatrix}. \quad (1.18)$$

The inverse kinematics solves

$$\boldsymbol{v} = Q_1 Q_3 \boldsymbol{v}_5. \quad (1.19)$$

The end-effector vector in frame 5 is

$$\boldsymbol{v}_5 = \begin{bmatrix} \sin \beta_2 \\ \sin \beta_1 \cos \beta_2 \\ \cos \beta_1 \cos \beta_2 \end{bmatrix}. \quad (1.20)$$

For this robot,

$$\beta_1 = 0, \quad \beta_2 = 75^\circ. \quad (1.21)$$

The inverse kinematics can produce multiple mathematical branches. Therefore, the solution is selected using physical joint limits and continuity with the previous solution.

For the special line

$$\phi_1 = 0, \quad (1.22)$$

the physically consistent branch is

$$\theta_1 = 0, \quad \theta_2 = \phi_2, \quad \theta_3 = 15^\circ + \theta_2. \quad (1.23)$$

This branch ensures consistency between inverse kinematics and direct kinematics when the mechanism passes through the $\theta_1 = 0$ line.

1.4 Initial Condition Correction

When the task-space trajectory starts from a nonzero pose, the initial joint position is also nonzero. Therefore, the plant must be initialized using the IK solution of the initial task-space pose.

If

$$\phi(0) = \phi_0, \quad (1.24)$$

then

$$\theta_j(0) = IK(\phi_0). \quad (1.25)$$

The joint-side initial condition is

$$\theta_{j,0} = \begin{bmatrix} \theta_{10} \\ \theta_{20} \end{bmatrix}. \quad (1.26)$$

Since the dynamics are modeled on the motor side, this initial condition is converted to motor side using the gearbox ratio:

$$\theta_{m,0} = N\theta_{j,0}. \quad (1.27)$$

The motor velocity initial condition is

$$\dot{\theta}_{m,0} = N\dot{\theta}_{j,0}. \quad (1.28)$$

Since the trajectory starts from rest,

$$\dot{\theta}_{j,0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad (1.29)$$

therefore,

$$\dot{\boldsymbol{\theta}}_{m,0} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}. \quad (1.30)$$

In Simulink, the motor-side position integrator is initialized with

$$\boldsymbol{\theta}_{m,0}. \quad (1.31)$$

The motor-side velocity integrator is initialized with

$$\dot{\boldsymbol{\theta}}_{m,0}. \quad (1.32)$$

This avoids the initial transient that occurs when the plant starts from zero while the desired trajectory starts from a nonzero configuration.

1.5 Jacobian Velocity Relation

The velocity relation is obtained by differentiating the geometric constraints. It is written as

$$K\dot{\boldsymbol{\theta}}_j = J\dot{\boldsymbol{\phi}}. \quad (1.33)$$

Therefore,

$$\dot{\boldsymbol{\theta}}_j = K^{-1}J\dot{\boldsymbol{\phi}}. \quad (1.34)$$

The matrix K is

$$K = \begin{bmatrix} k_{11} & 0 \\ 0 & k_{22} \end{bmatrix}. \quad (1.35)$$

The first diagonal term is

$$k_{11} = \mathbf{v}_1 \cdot (\mathbf{e}_1^1 \times \mathbf{e}_3^1). \quad (1.36)$$

The second diagonal term is

$$k_{22} = \mathbf{e}_5^1 \cdot (\mathbf{e}_2^1 \times \mathbf{e}_4^1). \quad (1.37)$$

The matrix J is obtained from

$$J = J_0 S, \quad (1.38)$$

where

$$J_0 = \begin{bmatrix} (\mathbf{e}_3^1 \times \mathbf{v}_1)^T \\ (\mathbf{e}_4^1 \times \mathbf{e}_5^1)^T \end{bmatrix}. \quad (1.39)$$

The angular velocity mapping is

$$\boldsymbol{\omega}_1 = S\dot{\boldsymbol{\phi}}. \quad (1.40)$$

Using the azimuth-tilt convention,

$$S = \begin{bmatrix} 0 & \sin \phi_1 \\ 0 & -\cos \phi_1 \\ 1 & 0 \end{bmatrix}. \quad (1.41)$$

Therefore,

$$\boldsymbol{\omega}_1 = \begin{bmatrix} 0 & \sin \phi_1 \\ 0 & -\cos \phi_1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{\phi}_1 \\ \dot{\phi}_2 \end{bmatrix}. \quad (1.42)$$

Therefore, the desired joint-side motor velocity is

$$\dot{\boldsymbol{\theta}}_{j,d} = K^{-1} J \dot{\boldsymbol{\phi}}_d. \quad (1.43)$$

1.6 Second-Order Jacobian Relation

The second-order Jacobian relation is used to compute the desired joint-side acceleration

$$\ddot{\boldsymbol{\theta}}_{j,d} = \begin{bmatrix} \ddot{\theta}_{1d} \\ \ddot{\theta}_{2d} \end{bmatrix}. \quad (1.44)$$

The acceleration equations are obtained by differentiating the constraint equations twice.

The first geometric constraint is

$$\mathbf{e}_3^1 \cdot \mathbf{v}_1 = 0. \quad (1.45)$$

Differentiating once gives

$$\dot{\mathbf{e}}_3^1 \cdot \mathbf{v}_1 + \mathbf{e}_3^1 \cdot \dot{\mathbf{v}}_1 = 0. \quad (1.46)$$

Differentiating again gives

$$\ddot{\mathbf{e}}_3^1 \cdot \mathbf{v}_1 + 2\dot{\mathbf{e}}_3^1 \cdot \dot{\mathbf{v}}_1 + \mathbf{e}_3^1 \cdot \ddot{\mathbf{v}}_1 = 0. \quad (1.47)$$

The second geometric constraint is

$$\mathbf{e}_4^1 \cdot \mathbf{e}_5^1 = 0. \quad (1.48)$$

Differentiating once gives

$$\dot{\mathbf{e}}_4^1 \cdot \mathbf{e}_5^1 + \mathbf{e}_4^1 \cdot \dot{\mathbf{e}}_5^1 = 0. \quad (1.49)$$

Differentiating again gives

$$\ddot{\mathbf{e}}_4^1 \cdot \mathbf{e}_5^1 + 2\dot{\mathbf{e}}_4^1 \cdot \dot{\mathbf{e}}_5^1 + \mathbf{e}_4^1 \cdot \ddot{\mathbf{e}}_5^1 = 0. \quad (1.50)$$

The vector derivatives are

$$\dot{\mathbf{e}}_3^1 = \dot{\theta}_1 (\mathbf{e}_1^1 \times \mathbf{e}_3^1), \quad (1.51)$$

$$\ddot{\mathbf{e}}_3^1 = \ddot{\theta}_1 (\mathbf{e}_1^1 \times \mathbf{e}_3^1) + \dot{\theta}_1 (\mathbf{e}_1^1 \times \dot{\mathbf{e}}_3^1). \quad (1.52)$$

Similarly,

$$\dot{\mathbf{e}}_4^1 = \dot{\theta}_2 (\mathbf{e}_2^1 \times \mathbf{e}_4^1), \quad (1.53)$$

$$\ddot{\mathbf{e}}_4^1 = \ddot{\theta}_2 (\mathbf{e}_2^1 \times \mathbf{e}_4^1) + \dot{\theta}_2 (\mathbf{e}_2^1 \times \dot{\mathbf{e}}_4^1). \quad (1.54)$$

The platform-vector derivatives are

$$\dot{\mathbf{v}}_1 = \boldsymbol{\omega}_1 \times \mathbf{v}_1, \quad (1.55)$$

$$\ddot{\mathbf{v}}_1 = \dot{\boldsymbol{\omega}}_1 \times \mathbf{v}_1 + \boldsymbol{\omega}_1 \times \dot{\mathbf{v}}_1. \quad (1.56)$$

For the passive axis,

$$\dot{\mathbf{e}}_5^1 = \boldsymbol{\omega}_1 \times \mathbf{e}_5^1, \quad (1.57)$$

$$\ddot{\mathbf{e}}_5^1 = \dot{\boldsymbol{\omega}}_1 \times \mathbf{e}_5^1 + \boldsymbol{\omega}_1 \times \dot{\mathbf{e}}_5^1. \quad (1.58)$$

The angular acceleration is

$$\dot{\boldsymbol{\omega}}_1 = S\ddot{\boldsymbol{\phi}} + \dot{S}\dot{\boldsymbol{\phi}}. \quad (1.59)$$

Using the first acceleration constraint, the first joint acceleration is

$$\ddot{\theta}_1 = -\frac{\dot{\theta}_1 (\mathbf{e}_1^1 \times \dot{\mathbf{e}}_3^1) \cdot \mathbf{v}_1 + 2\dot{\mathbf{e}}_3^1 \cdot \dot{\mathbf{v}}_1 + \mathbf{e}_3^1 \cdot \ddot{\mathbf{v}}_1}{\mathbf{v}_1 \cdot (\mathbf{e}_1^1 \times \mathbf{e}_3^1)}. \quad (1.60)$$

Using the second acceleration constraint, the second joint acceleration is

$$\ddot{\theta}_2 = -\frac{\dot{\theta}_2 (\mathbf{e}_2^1 \times \dot{\mathbf{e}}_4^1) \cdot \mathbf{e}_5^1 + 2\dot{\mathbf{e}}_4^1 \cdot \dot{\mathbf{e}}_5^1 + \mathbf{e}_4^1 \cdot \ddot{\mathbf{e}}_5^1}{\mathbf{e}_5^1 \cdot (\mathbf{e}_2^1 \times \mathbf{e}_4^1)}. \quad (1.61)$$

Therefore,

$$\ddot{\boldsymbol{\theta}}_{j,d} = \begin{bmatrix} \ddot{\theta}_{1d} \\ \ddot{\theta}_{2d} \end{bmatrix}. \quad (1.62)$$

1.7 Motor-Side Conversion

The inverse kinematics, Jacobian velocity relation, and second-order Jacobian relation give joint-side references:

$$\boldsymbol{\theta}_{j,d}, \quad \dot{\boldsymbol{\theta}}_{j,d}, \quad \ddot{\boldsymbol{\theta}}_{j,d}. \quad (1.63)$$

However, the control and dynamics are expressed on the motor side. The gearbox relation is

$$\boldsymbol{\theta}_m = N\boldsymbol{\theta}_j. \quad (1.64)$$

For this robot,

$$N = 70. \quad (1.65)$$

Therefore,

$$\boldsymbol{\theta}_{m,d} = N\boldsymbol{\theta}_{j,d}, \quad (1.66)$$

$$\dot{\boldsymbol{\theta}}_{m,d} = N\dot{\boldsymbol{\theta}}_{j,d}, \quad (1.67)$$

$$\ddot{\boldsymbol{\theta}}_{m,d} = N\ddot{\boldsymbol{\theta}}_{j,d}. \quad (1.68)$$

These motor-side references are used by the controller.

After the motor-side plant is integrated, the actual motor angle is converted back to joint side using

$$\boldsymbol{\theta}_j = \frac{\boldsymbol{\theta}_m}{N}, \quad (1.69)$$

$$\dot{\theta}_j = \frac{\dot{\theta}_m}{N}. \quad (1.70)$$

The direct kinematics block uses joint-side angles, not motor-side angles.

1.8 Simplified Decoupled Computed-Torque Model

A full robot dynamic model can be written as

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau. \quad (1.71)$$

This model includes inertia coupling, Coriolis effects, centrifugal effects, gravity, friction, and the interaction between different robot links.

For the dynamic implementation, the model is simplified by treating each motor axis independently. This gives a decoupled motor-side model.

For one rotational actuator, the basic rotational dynamics equation is

$$\tau_m = J_m \ddot{\theta}_m. \quad (1.72)$$

This is the rotational equivalent of Newton's law

$$F = ma. \quad (1.73)$$

For rotational motion,

$$F \rightarrow \tau, \quad m \rightarrow J, \quad a \rightarrow \ddot{\theta}. \quad (1.74)$$

Therefore,

$$\tau = J\ddot{\theta}. \quad (1.75)$$

For the two motors of the robot,

$$\tau_{m1} = J_{m1} \ddot{\theta}_{m1}, \quad (1.76)$$

$$\tau_{m2} = J_{m2} \ddot{\theta}_{m2}. \quad (1.77)$$

This means that motor 1 torque mainly produces motor 1 acceleration, and motor 2 torque mainly produces motor 2 acceleration.

The term computed torque means that the controller computes the motor torque required to generate a desired corrected acceleration.

The motor-side tracking error is

$$\mathbf{e}_m = \boldsymbol{\theta}_{m,d} - \boldsymbol{\theta}_m. \quad (1.78)$$

The motor-side velocity error is

$$\dot{\mathbf{e}}_m = \dot{\boldsymbol{\theta}}_{m,d} - \dot{\boldsymbol{\theta}}_m. \quad (1.79)$$

The integral error is

$$\mathbf{e}_{I,m} = \int_0^t (\boldsymbol{\theta}_{m,d} - \boldsymbol{\theta}_m) dt. \quad (1.80)$$

A corrected motor acceleration command is constructed as

$$\ddot{\boldsymbol{\theta}}_{m,cmd} = \ddot{\boldsymbol{\theta}}_{m,d} + K_d \dot{\mathbf{e}}_m + K_p \mathbf{e}_m + K_i \mathbf{e}_{I,m}. \quad (1.81)$$

The motor torque is then computed as

$$\boldsymbol{\tau}_m = J_m \ddot{\boldsymbol{\theta}}_{m,cmd}. \quad (1.82)$$

For motor 1,

$$\tau_{m1} = J_{m1} \left[\ddot{\theta}_{m1,d} + K_{d1} (\dot{\theta}_{m1,d} - \dot{\theta}_{m1}) + K_{p1} (\theta_{m1,d} - \theta_{m1}) + K_{i1} \int_0^t (\theta_{m1,d} - \theta_{m1}) dt \right]. \quad (1.83)$$

For motor 2,

$$\tau_{m2} = J_{m2} \left[\ddot{\theta}_{m2,d} + K_{d2} (\dot{\theta}_{m2,d} - \dot{\theta}_{m2}) + K_{p2} (\theta_{m2,d} - \theta_{m2}) + K_{i2} \int_0^t (\theta_{m2,d} - \theta_{m2}) dt \right]. \quad (1.84)$$

1.9 Controller Gain Calculation

The controller gains are selected so that the closed-loop tracking error behaves like a stable third-order system.

The motor-side computed-torque control law is

$$\tau_m = J_m \left[\ddot{\theta}_{m,d} + K_d (\dot{\theta}_{m,d} - \dot{\theta}_m) + K_p (\theta_{m,d} - \theta_m) + K_i \int_0^t (\theta_{m,d} - \theta_m) dt \right]. \quad (1.85)$$

The simplified motor dynamics is

$$\tau_m = J_m \ddot{\theta}_m. \quad (1.86)$$

Substituting the controller into the dynamics gives

$$J_m \ddot{\theta}_m = J_m \left[\ddot{\theta}_{m,d} + K_d(\dot{\theta}_{m,d} - \dot{\theta}_m) + K_p(\theta_{m,d} - \theta_m) + K_i \int_0^t (\theta_{m,d} - \theta_m) dt \right]. \quad (1.87)$$

Canceling J_m ,

$$\ddot{\theta}_m = \ddot{\theta}_{m,d} + K_d \dot{e}_m + K_p e_m + K_i \int_0^t e_m dt. \quad (1.88)$$

Using

$$e_m = \theta_{m,d} - \theta_m, \quad (1.89)$$

$$\dot{e}_m = \dot{\theta}_{m,d} - \dot{\theta}_m, \quad (1.90)$$

$$\ddot{e}_m = \ddot{\theta}_{m,d} - \ddot{\theta}_m, \quad (1.91)$$

the error equation becomes

$$\ddot{e}_m + K_d \dot{e}_m + K_p e_m + K_i \int_0^t e_m dt = 0. \quad (1.92)$$

Differentiating once removes the integral term:

$$\ddot{\ddot{e}}_m + K_d \ddot{e}_m + K_p \dot{e}_m + K_i e_m = 0. \quad (1.93)$$

The characteristic polynomial is therefore

$$s^3 + K_d s^2 + K_p s + K_i. \quad (1.94)$$

To obtain a stable third-order response with three identical poles at

$$s = -W_n, \quad (1.95)$$

the desired characteristic polynomial is chosen as

$$(s + W_n)^3. \quad (1.96)$$

Expanding,

$$(s + W_n)^3 = s^3 + 3W_n s^2 + 3W_n^2 s + W_n^3. \quad (1.97)$$

By matching coefficients,

$$K_d = 3W_n, \quad (1.98)$$

$$K_p = 3W_n^2, \quad (1.99)$$

$$K_i = W_n^3. \quad (1.100)$$

The cutoff frequency W_c is chosen as a tuning parameter. For the Beckhoff axes,

$$W_c = 150. \quad (1.101)$$

The natural frequency is defined as

$$W_n = 2W_c. \quad (1.102)$$

Therefore,

$$W_n = 2(150) = 300. \quad (1.103)$$

The gains become

$$K_d = 3(300) = 900, \quad (1.104)$$

$$K_p = 3(300)^2 = 270000, \quad (1.105)$$

$$K_i = 300^3 = 27000000. \quad (1.106)$$

Thus,

$$K_{p1} = K_{p2} = 270000, \quad (1.107)$$

$$K_{d1} = K_{d2} = 900, \quad (1.108)$$

$$K_{i1} = K_{i2} = 27000000. \quad (1.109)$$

1.10 Motor-Side Dynamics

The motor dynamics are written on the motor side as

$$J_m \ddot{\boldsymbol{\theta}}_m = \boldsymbol{\tau}_m. \quad (1.110)$$

Therefore,

$$\ddot{\boldsymbol{\theta}}_m = J_m^{-1} \boldsymbol{\tau}_m. \quad (1.111)$$

For the two motors,

$$\ddot{\theta}_{m1} = \frac{\tau_{m1}}{J_{m1}}, \quad (1.112)$$

$$\ddot{\theta}_{m2} = \frac{\tau_{m2}}{J_{m2}}. \quad (1.113)$$

The motor acceleration is integrated once to obtain motor velocity:

$$\ddot{\boldsymbol{\theta}}_m \rightarrow \dot{\boldsymbol{\theta}}_m. \quad (1.114)$$

It is integrated again to obtain motor position:

$$\dot{\boldsymbol{\theta}}_m \rightarrow \boldsymbol{\theta}_m. \quad (1.115)$$

The motor-side states are then converted back to joint-side states:

$$\boldsymbol{\theta}_j = \frac{\boldsymbol{\theta}_m}{N}, \quad (1.116)$$

$$\dot{\boldsymbol{\theta}}_j = \frac{\dot{\boldsymbol{\theta}}_m}{N}. \quad (1.117)$$

1.11 Motor-Side Equivalent Inertia

The motor-side equivalent inertia is mainly composed of the motor rotor inertia, brake inertia, and gear inertia:

$$J_m = J_{rotor} + J_{brake} + J_{gear}. \quad (1.118)$$

Using the motor data,

$$J_{rotor} = 38 \times 10^{-6} \text{ kg m}^2, \quad (1.119)$$

$$J_{brake} = 7 \times 10^{-6} \text{ kg m}^2, \quad (1.120)$$

$$J_{gear} = 17 \times 10^{-6} \text{ kg m}^2. \quad (1.121)$$

Therefore,

$$J_m = 38 \times 10^{-6} + 7 \times 10^{-6} + 17 \times 10^{-6}. \quad (1.122)$$

$$J_m = 62 \times 10^{-6} \text{ kg m}^2. \quad (1.123)$$

Thus,

$$J_{m1} = J_{m2} = 62 \times 10^{-6} \text{ kg m}^2. \quad (1.124)$$

The gearbox relation is

$$\theta_m = N\theta_j. \quad (1.125)$$

Therefore, any joint-side inertia reflected to the motor side is divided by the square of the gear ratio:

$$J_{j \rightarrow m} = \frac{J_j}{N^2}. \quad (1.126)$$

For this robot,

$$N = 70, \quad (1.127)$$

so

$$N^2 = 4900. \quad (1.128)$$

Therefore,

$$J_{j \rightarrow m} = \frac{J_j}{4900}. \quad (1.129)$$

This means that the joint-side load inertia has a reduced effect when observed from the motor shaft. Consequently, the motor-side dynamics are mainly governed by the motor inertia, brake inertia, and gear inertia.

1.12 Torque Saturation

The motor-side torque command is limited by the maximum motor torque:

$$|\tau_m| \leq 1 \text{ Nm.} \quad (1.130)$$

For the two motors,

$$|\tau_{m1}| \leq 1 \text{ Nm,} \quad (1.131)$$

$$|\tau_{m2}| \leq 1 \text{ Nm.} \quad (1.132)$$

The corresponding joint-side torque is obtained from

$$\tau_j = \eta N \tau_m, \quad (1.133)$$

where

$$\eta = 0.94, \quad N = 70. \quad (1.134)$$

Therefore, for

$$\tau_m = 1 \text{ Nm,} \quad (1.135)$$

the joint-side torque is

$$\tau_j = 0.94 \times 70 \times 1. \quad (1.136)$$

$$\tau_j = 65.8 \text{ Nm.} \quad (1.137)$$

Thus, the motor torque limit of 1 Nm corresponds to a joint-side torque limit of 65.8 Nm.

1.13 Direct Kinematics Feedback

The direct kinematics block uses joint-side angles. Therefore, after the motor-side plant, the actual motor angle is converted back to joint side:

$$\theta_j = \frac{\theta_m}{N}. \quad (1.138)$$

The direct kinematics block receives

$$\boldsymbol{\theta}_j = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}. \quad (1.139)$$

The DK block computes the passive angle θ_3 , then calculates

$$\mathbf{v}_1 = Q_1 Q_3 \mathbf{v}_5. \quad (1.140)$$

The final-frame direction is

$$\mathbf{d} = \begin{bmatrix} -\mathbf{v}_1(2) \\ \mathbf{v}_1(1) \\ \mathbf{v}_1(3) \end{bmatrix}. \quad (1.141)$$

The task-space orientation is reconstructed as

$$\phi_1 = \text{atan2}(-d_x, d_y), \quad (1.142)$$

$$\phi_2 = \sin^{-1}(d_z). \quad (1.143)$$

Therefore,

$$\boldsymbol{\phi}_{act} = \begin{bmatrix} \phi_{1act} \\ \phi_{2act} \end{bmatrix}. \quad (1.144)$$

This is compared with the desired trajectory

$$\boldsymbol{\phi}_d. \quad (1.145)$$

1.14 Signal Units

Internally, Simulink uses radians for angular quantities.

The task-space orientation has units

$$\phi_1, \phi_2 : \text{rad}. \quad (1.146)$$

If a gain $180/\pi$ is applied for plotting, the displayed unit is degrees.

The joint-side angular quantities are

$$\theta_j : \text{rad}, \quad (1.147)$$

$$\dot{\theta}_j : \text{rad/s}, \quad (1.148)$$

$$\ddot{\theta}_j : \text{rad/s}^2. \quad (1.149)$$

The motor-side angular quantities are

$$\theta_m : \text{rad}, \quad (1.150)$$

$$\dot{\theta}_m : \text{rad/s}, \quad (1.151)$$

$$\ddot{\theta}_m : \text{rad/s}^2. \quad (1.152)$$

The torque command is

$$\tau_m : \text{Nm}. \quad (1.153)$$

The conversion between motor and joint side is

$$\theta_m = 70\theta_j, \quad (1.154)$$

$$\dot{\theta}_m = 70\dot{\theta}_j, \quad (1.155)$$

$$\ddot{\theta}_m = 70\ddot{\theta}_j. \quad (1.156)$$

For plotting, θ_m can appear very large in degrees because the motor rotates 70 times more than the joint. For example,

$$\theta_j = 30^\circ \quad (1.157)$$

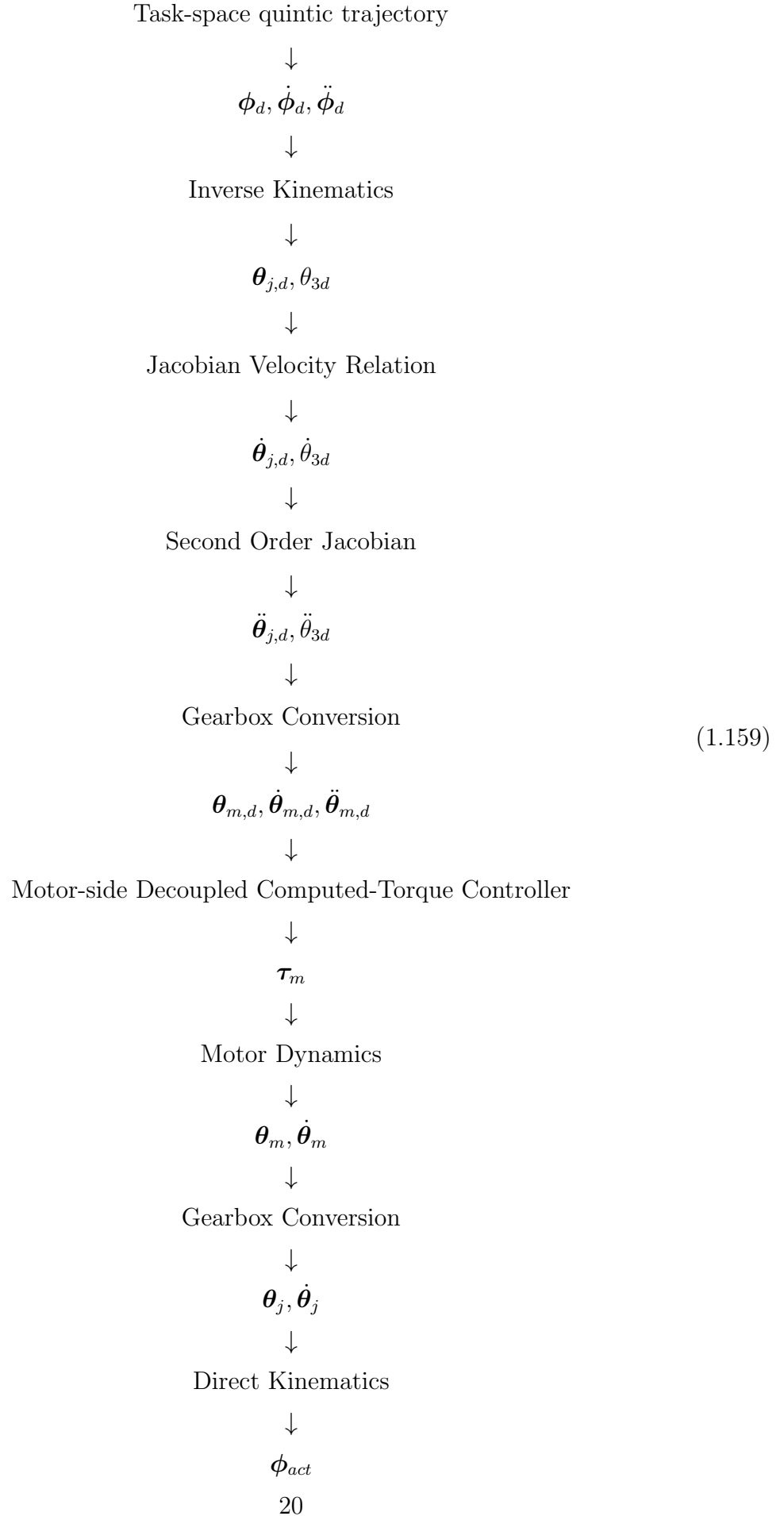
corresponds to

$$\theta_m = 70 \times 30^\circ = 2100^\circ. \quad (1.158)$$

Therefore, motor-side angle plots may show values in thousands of degrees, while joint-side plots remain in the physical joint range.

1.15 Final Simulink Structure

The final Simulink structure is



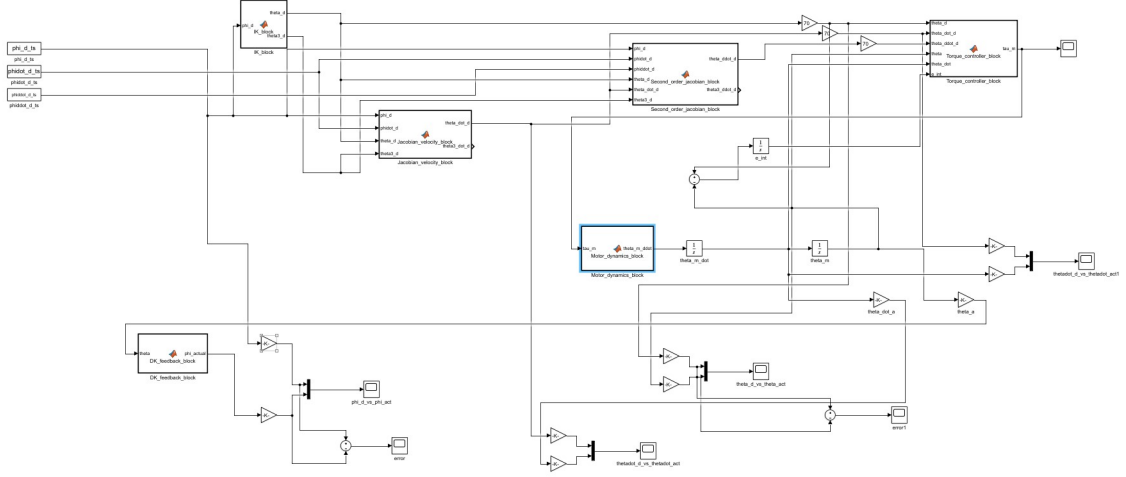


Figure 1.1: Complete Simulink layout for task-space trajectory tracking with motor-side decoupled computed-torque control.

1.16 Simulation Result Interpretation

The simulation results are shown in Fig. 1.3. The figure contains the task-space tracking response, the motor-side velocity tracking response, and the motor-side position tracking response.

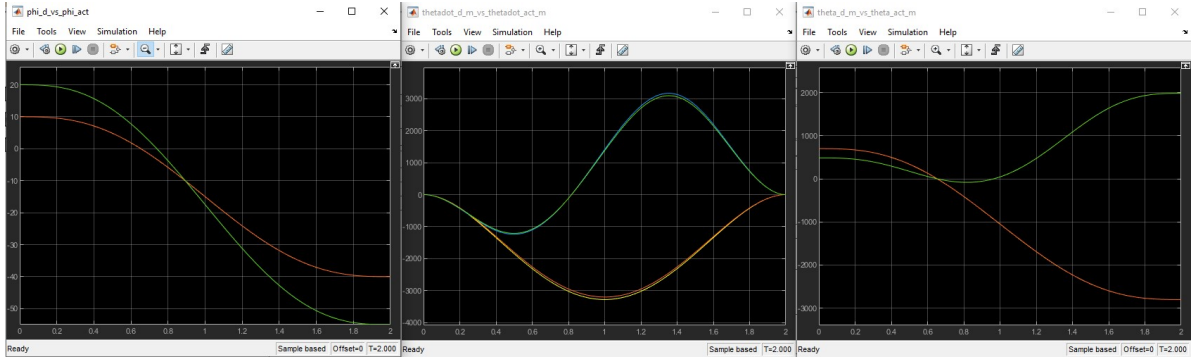


Figure 1.2: Simulation results for task-space tracking and motor-side actuator response. The left plot shows the desired and actual task-space orientation, the middle plot shows the desired and actual motor-side velocities, and the right plot shows the desired and actual motor-side positions.

The task-space response compares the desired orientation

$$\phi_d = \begin{bmatrix} \phi_{1d} \\ \phi_{2d} \end{bmatrix} \quad (1.160)$$

with the actual orientation obtained from the direct kinematics,

$$\phi_{act} = \begin{bmatrix} \phi_{1act} \\ \phi_{2act} \end{bmatrix}. \quad (1.161)$$

The desired and actual task-space trajectories are very close, which shows that the direct kinematics feedback follows the desired trajectory correctly:

$$\phi_{act} \approx \phi_d. \quad (1.162)$$

The task-space trajectory gives smooth position, velocity, and acceleration profiles. Since the final angles are smaller than the initial angles for this trajectory, the corresponding velocities become negative during the motion. This indicates that the robot is moving in the negative task-space direction.

The quintic trajectory satisfies the boundary conditions

$$\dot{\phi}(0) = 0, \quad \dot{\phi}(t_f) = 0, \quad (1.163)$$

and

$$\ddot{\phi}(0) = 0, \quad \ddot{\phi}(t_f) = 0. \quad (1.164)$$

The middle plot in Fig. 1.3 shows the motor-side velocity tracking. The desired motor velocity is obtained from the joint-side velocity using the gearbox ratio,

$$\dot{\theta}_{m,d} = N\dot{\theta}_{j,d}. \quad (1.165)$$

The actual motor velocity is obtained by integrating the motor acceleration in the motor dynamics block. The close overlap between desired and actual motor velocity confirms that the motor-side controller tracks the velocity reference correctly:

$$\dot{\theta}_m \approx \dot{\theta}_{m,d}. \quad (1.166)$$

The right plot in Fig. 1.3 shows the motor-side position tracking. Since the motor rotates faster than the joint by the gearbox ratio,

$$\theta_m = N\theta_j, \quad (1.167)$$

the motor-side angle values are much larger than the joint-side angles. For example, with

$$N = 70, \quad (1.168)$$

a joint rotation of 30° corresponds to a motor rotation of

$$\theta_m = 70 \times 30^\circ = 2100^\circ. \quad (1.169)$$

Therefore, motor-side position plots can show values in thousands of degrees. This is expected and does not indicate an error. The important point is that the desired and actual motor-side position curves follow each other:

$$\boldsymbol{\theta}_m \approx \boldsymbol{\theta}_{m,d}. \quad (1.170)$$

The motor torque response remains within the motor torque limit:

$$|\tau_m| \leq 1 \text{ Nm}. \quad (1.171)$$

If the maximum motor torque is approximately

$$0.55 \text{ Nm}, \quad (1.172)$$

then the corresponding joint-side torque is

$$\tau_j = \eta N \tau_m. \quad (1.173)$$

Using

$$\eta = 0.94, \quad N = 70, \quad (1.174)$$

we obtain

$$\tau_j = 0.94 \times 70 \times 0.55. \quad (1.175)$$

Therefore,

$$\tau_j \approx 36.2 \text{ Nm}. \quad (1.176)$$

This remains below the available joint-side torque limit:

$$65.8 \text{ Nm}. \quad (1.177)$$

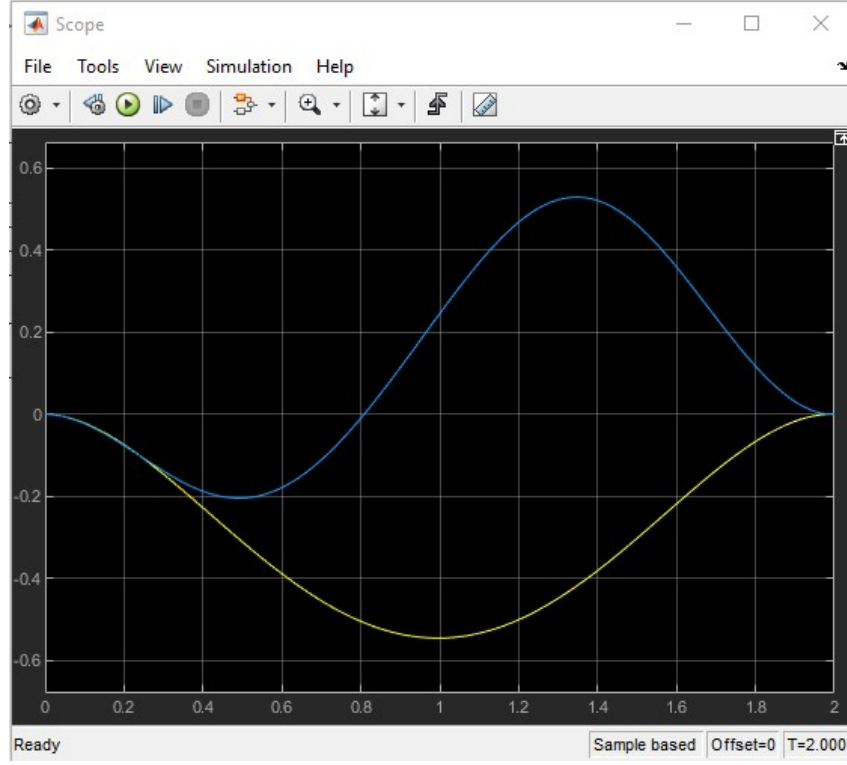


Figure 1.3: Torque produced by motor 1 and motor 2

Therefore, the trajectory can be tracked without actuator saturation.

Overall, the simulation confirms that the task-space trajectory generation, inverse kinematics, Jacobian velocity relation, second-order Jacobian relation, motor-side controller, motor dynamics, gearbox conversion, and direct kinematics feedback are working together correctly.

1.16.1 Motor-Side Acceleration Tracking

The desired and actual motor-side accelerations are shown in Fig. 1.4. The desired motor-side acceleration is obtained from the desired joint-side acceleration using the gearbox ratio:

$$\ddot{\theta}_{m,d} = N\ddot{\theta}_{j,d}. \quad (1.178)$$

Since the gearbox ratio is

$$N = 70, \quad (1.179)$$

the motor-side acceleration is much larger than the joint-side acceleration. This is expected because the motor rotates faster than the joint:

$$\theta_m = N\theta_j, \quad \dot{\theta}_m = N\dot{\theta}_j, \quad \ddot{\theta}_m = N\ddot{\theta}_j. \quad (1.180)$$

The actual motor-side acceleration is obtained from the motor dynamics block:

$$\ddot{\theta}_{m,act} = \frac{\tau_m}{J_m}. \quad (1.181)$$

The desired and actual acceleration curves do not overlap perfectly because the actual acceleration is generated by the computed-torque controller. The controller does not apply only the desired feedforward acceleration. It also adds correction terms to reduce the position and velocity tracking errors:

$$\ddot{\theta}_{m,act} = \ddot{\theta}_{m,d} + K_d (\dot{\theta}_{m,d} - \dot{\theta}_m) + K_p (\theta_{m,d} - \theta_m) + K_i \int_0^t (\theta_{m,d} - \theta_m) dt. \quad (1.182)$$

Therefore, the actual acceleration can differ from the desired acceleration even when the position and velocity tracking are good. Since acceleration is integrated to obtain velocity and position, these acceleration differences are smoothed by the integrators. The close overlap of the motor-side velocity and position curves confirms that the controller tracks the desired motion correctly.

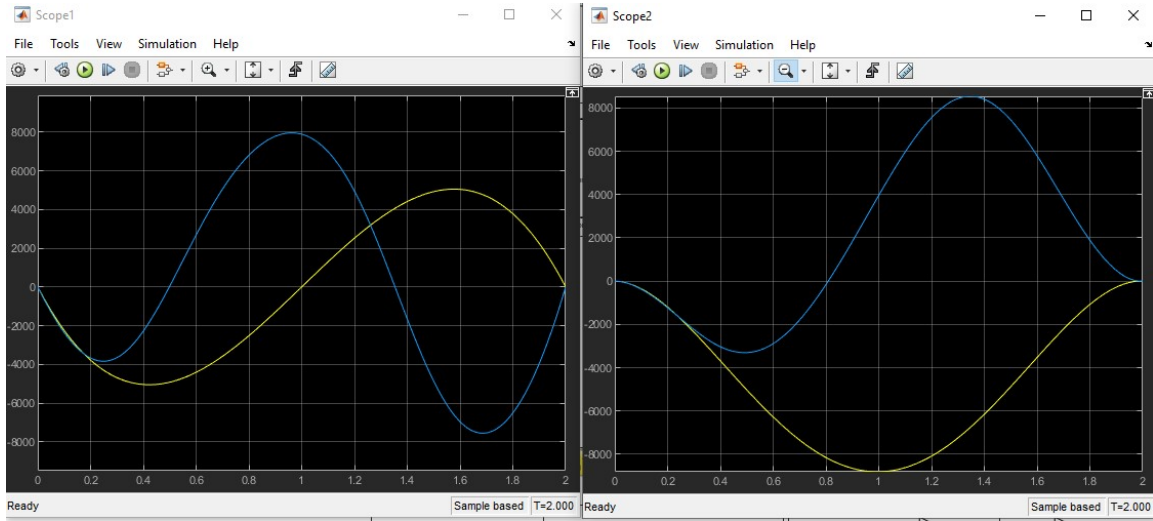


Figure 1.4: Desired and actual motor-side acceleration response for motors 1 and 2. The desired acceleration is obtained by multiplying the joint-side acceleration by the gearbox ratio $N = 70$, while the actual acceleration is obtained from the motor dynamics block.

1.17 Conclusion

A complete task-space trajectory tracking model was implemented in Simulink for the 2-DOF spherical robot. The trajectory is generated in task space using quintic polynomials. The inverse kinematics converts the task-space orientation into desired joint-side motor angles. The Jacobian velocity relation computes desired joint velocities, and the second-order Jacobian relation computes desired joint accelerations.

Since the actuator dynamics are modeled on the motor side, the joint-side references are converted into motor-side references using the gearbox ratio

$$N = 70. \quad (1.183)$$

The controller is implemented as a motor-side decoupled computed-torque controller. Each motor is treated as an independent rotational inertia system:

$$\tau_{m1} = J_{m1}\ddot{\theta}_{m1}, \quad (1.184)$$

$$\tau_{m2} = J_{m2}\ddot{\theta}_{m2}. \quad (1.185)$$

The computed-torque control law calculates the torque required to generate a corrected motor acceleration command. The corrected acceleration includes the desired motor acceleration and PID tracking error terms.

The motor dynamics are modeled using

$$\ddot{\theta}_m = \frac{\tau_m}{J_m}. \quad (1.186)$$

The motor-side states are integrated to obtain motor velocity and motor position. These states are then converted back to joint side using

$$\theta_j = \frac{\theta_m}{N}. \quad (1.187)$$

The direct kinematics block uses the joint-side angles to reconstruct the actual task-space orientation.

The simulation results show that the desired and actual task-space trajectories match closely:

$$\phi_{act} \approx \phi_d. \quad (1.188)$$

The motor torque remains within the allowable torque limit, confirming that the selected trajectory is feasible for the simplified motor-side dynamic model.